

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-KC-4501-05 Sheet 1 of 1
OPL Title	Crosshead Vibration Monitoring (VSHH-4505)	Discipline	Instrument	
Equipment	KC-4501 - RECYCLE GAS COMPRESSOR	Area / Unit	4500 - RECYCLE GAS SYSTEM	
Related Interlock	SEQ-4501 (SIL 2)	P&ID Ref	TJC-LLD-PID-4501	
Classification:	☐ Basic Knowledge	☒ Improvement	☐ Trouble Case	Aspect: P-Q-C-D- S -M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **crosshead vibration monitoring (vshh-4505)** for KC-4501 (RECYCLE GAS COMPRESSOR) in the RECYCLE GAS SYSTEM. A rising crosshead vibration trend warns of loose running gear before it trips or fails. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Ethylene/Hexane/H2** at up to 16 barg / 150 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Trend VT-4501 crosshead vibration on the daily route	Loop isolated / under permit
2	Alarm at 8 mm/s and trip at 11 mm/s (VSHH-4505)	Reference standard traceable
3	Correlate any rise with TE-4507 main bearing temperature	As-found & as-left recorded
4	Investigate rod-load reversals and crosshead pin clearance	Trip / control action verified
5	Escalate a sudden step-change to reliability immediately	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Proof test of lube oil low-low trip	Set-point within tolerance	Recalibrated PSLL-4504 to 1.5 barg, function tested
Seal gas flow FT-4506 low, buffer pressure unstable	N2 buffer regulator diaphragm failed	Replaced seal gas regulator, re-established buffer dP
Proof test of KO drum high-high level trip	Float healthy, minor set-point trim	Verified LSHH-4510 float and trip action

6. KEY LEARNING POINTS

- A rising crosshead vibration trend warns of loose running gear before it trips or fails.
- Respect the interlock SEQ-4501: e.g. PSLL-4504 trips at < 1.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-4501 before working on KC-4501.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

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